Research Analysis Report

Research Topic: Multi-Agent System

Total Papers: 5

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Executive Summary

This report provides a technical overview of 'Multi-Agent System' including methodologies, research gaps and experiment planning.

Literature Review

Introduction: This literature review summarizes findings from 5 research papers related to the selected topic. It highlights research trends, methodologies, key contributions, future directions, and overall observations.

Paper Summaries

• {'title': 'Magentic-One: A Generalist Multi-Agent System for Solving Complex Tasks', 'year': 2024, 'summary': "Modern AI agents, driven by advances in large foundation models, promise to enhance our productivity and transform our lives by augmenting our knowledge and capabilities. To achieve this vision, AI agents must effectively plan, perform multi-step reasoning and actions, respond to novel observations, and recover from errors, to successfully complete complex tasks across a wide range of scenarios. In this work, we introduce Magentic-One, a high-performing open-source agentic system for solving such tasks. Magentic-One uses a multi-agent architecture where a lead agent, the Orchestrator, plans, tracks progress, and re-plans to recover from errors. Throughout task execution, the Orchestrator directs other specialized agents to perform tasks as needed, such as operating a web browser, navigating local files, or writing and executing Python code. We show that Magentic-One achieves statistically competitive performance to the state-of-the-art on three diverse and challenging agentic benchmarks: GAIA, AssistantBench, and WebArena. Magentic-One achieves these results without modification to core agent capabilities or to how they collaborate, demonstrating progress towards generalist agentic systems. Moreover, Magentic-One's modular design allows agents to be added or removed from the team without additional prompt tuning or training, easing development and making it extensible to future scenarios. We provide an open-source implementation of"}

• {'title': 'PsySafe: A Comprehensive Framework for Psychological-based Attack, Defense, and Evaluation of Multi-agent System Safety', 'year': 2024, 'summary': "Multi-agent systems, when enhanced with Large Language Models (LLMs), exhibit profound capabilities in collective intelligence. However, the potential misuse of this intelligence for malicious purposes presents significant risks. To date, comprehensive research on the safety issues associated with multi-agent systems remains limited. In this paper, we explore these concerns through the innovative lens of agent psychology, revealing that the dark psychological states of agents constitute a significant threat to safety. To tackle these concerns, we propose a comprehensive framework (PsySafe) grounded in agent psychology, focusing on three key areas: firstly, identifying how dark personality traits in agents can lead to risky behaviors; secondly, evaluating the safety of multi-agent systems from the psychological and behavioral perspectives, and thirdly, devising effective strategies to mitigate these risks. Our experiments reveal several intriguing phenomena, such as the collective dangerous behaviors among agents, agents' self-reflection when engaging in dangerous behavior, and the correlation between agents' psychological assessments and dangerous behaviors. We anticipate that our framework and observations will provide valuable insights for further research into the safety of multi-agent systems. We will make our data and code publicly accessible at https://github.com/AI4Good24/PsySafe."}

• {'title': 'Towards smart factory for industry 4.0: a self-organized multi-agent system with big data based feedback and coordination', 'year': 2016, 'summary': 'Abstract not available for analysis.'}

• {'title': 'MIRIX: Multi-Agent Memory System for LLM-Based Agents', 'year': 2025, 'summary': "Although memory capabilities of AI agents are gaining increasing attention, existing solutions remain fundamentally limited. Most rely on flat, narrowly scoped memory components, constraining their ability to personalize, abstract, and reliably recall user-specific information over time. To this end, we introduce MIRIX, a modular, multi-agent memory system that redefines the future of AI memory by solving the field's most critical challenge: enabling language models to truly remember. Unlike prior approaches, MIRIX transcends text to embrace rich visual and multimodal experiences, making memory genuinely useful in real-world scenarios. MIRIX consists of six distinct, carefully structured memory types: Core, Episodic, Semantic, Procedural, Resource Memory, and Knowledge Vault, coupled with a multi-agent framework that dynamically controls and coordinates updates and retrieval. This design enables agents to persist, reason over, and accurately retrieve diverse, long-term user data at scale. We validate MIRIX in two demanding settings. First, on ScreenshotVQA, a challenging multimodal benchmark comprising nearly 20,000 high-resolution computer screenshots per sequence, requiring deep contextual understanding and where no existing memory systems can be applied, MIRIX achieves 35% higher accuracy than the RAG baseline while reducing storage requirements by 99.9%. Second, on LOCOMO, a long-form conversation benchmark with single-modal textual input, MIRIX attains state-of-the-art performance"}

• {'title': 'A multi-agent system for automating scientific discovery', 'year': 2026, 'summary': 'Scientific discovery is driven by the iterative process of observation, hypothesis generation, experimentation and data analysis. Despite recent advancements in applying artificial intelligence (AI) to biology, no system has yet automated all these stages1, 2–3. Here we introduce Robin, a multi-agent system capable of fully automating both hypothesis generation and data analysis for experimental biology. By integrating literature search agents with data analysis agents, Robin can generate hypotheses, propose experiments, interpret experimental results and generate updated hypotheses, achieving a semi-autonomous approach to scientific discovery. By applying this system, we were able to identify promising therapeutic candidates for dry age-related macular degeneration, the major cause of blindness in the developed world4,5. Robin proposed enhancing retinal pigment epithelium phagocytosis as a therapeutic strategy, and identified and confirmed in vitro efficacy for ripasudil and KL001. Ripasudil is a clinically used Rho kinase inhibitor that, to our knowledge, has never previously been proposed for the treatment of dry age-related macular degeneration. To elucidate the mechanism of ripasudil-induced upregulation of phagocytosis, Robin then proposed and analysed a follow-up RNA sequencing experiment, which revealed upregulation of ABCA1, which encodes a lipid efflux pump and represents a possible novel target. All hypotheses, experimental directions, data analyses and'}

Research Trends

• accessible

• accurately

• achieve

• age-related

• agent

• agentic

• behavior

- benchmark
- collective
- concern
- dangerous
- dark
- degeneration
- experiment
- experimental
- follow-up
- generation
- high-performing
- high-resolution
- hypothese
- hypothesis
- intelligence
- knowledge
- long-form
- long-term
- magentic-one
- memory
- mirix
- multi-agent
- multi-step
- multimodal
- open-source
- phagocytosis
- psychological
- psychology
- psysafe
- re-plan
- real-world
- reasoning
- ripasudil-induced
- risks
- robin
- safety
- self-reflection
- semi-autonomous

• single-modal

• state-of-the-art

• tasks

• user-specific

• vision

Research Gaps

• We anticipate that our framework and observations will provide valuable insights for further research into the safety of multi-agent systems.

Future Scope

• We anticipate that our framework and observations will provide valuable insights for further research into the safety of multi-agent systems.

Conclusion: Overall, the reviewed studies demonstrate significant progress in this research area while highlighting several opportunities for future work and further experimentation.

Methodology Comparison

Total Papers: 5

Comparison

• {'title': 'Magentic-One: A Generalist Multi-Agent System for Solving Complex Tasks', 'research_area': 'Multi-Agent Systems', 'research_problem': 'Modern AI agents, driven by advances in large foundation models, promise to enhance our productivity and transform our lives by augmenting our knowledge and capabilities.', 'methodology': ['Architecture', 'Experimental Evaluation'], 'keywords': ['agent', 'multi-agent', 'magentic-one', 'open-source', 'benchmark', 'knowledge', 'reasoning', 'vision', 'high-performing', 'multi-step', 'state-of-the-art', 're-plan', 'achieve', 'agentic', 'tasks'], 'year': 2024, 'citation_count': 251, 'quality_score': 80, 'quality_classification': 'Very Good'}

• {'title': 'PsySafe: A Comprehensive Framework for Psychological-based Attack, Defense, and Evaluation of Multi-agent System Safety', 'research_area': 'Multi-Agent Systems', 'research_problem': 'However, the potential misuse of this intelligence for malicious purposes presents significant risks.', 'methodology': ['Framework', 'Experimental Evaluation', 'Large Language Model'], 'keywords': ['multi-agent', 'agent', 'self-reflection', 'behavior', 'safety', 'psychological', 'dangerous', 'collective', 'intelligence', 'psychology', 'concern', 'dark', 'psysafe', 'risks', 'accessible'], 'year': 2024, 'citation_count': 98, 'quality_score': 70, 'quality_classification': 'Very Good'}

• {'title': 'Towards smart factory for industry 4.0: a self-organized multi-agent system with big data based feedback and coordination', 'research_area': 'Multi-Agent Systems', 'research_problem': 'Not Available', 'methodology': ['Not Identified'], 'keywords': [], 'year': 2016, 'citation_count': 1263, 'quality_score': 50, 'quality_classification': 'Good'}

• {'title': 'MIRIX: Multi-Agent Memory System for LLM-Based Agents', 'research_area': 'Retrieval-Augmented Generation', 'research_problem': 'Although memory capabilities of AI agents are gaining increasing attention, existing solutions remain fundamentally limited.', 'methodology': ['Framework', 'Experimental Evaluation', 'Retrieval-Augmented Generation'], 'keywords': ['memory', 'multi-agent', 'agent', 'benchmark', 'mirix', 'knowledge', 'high-resolution', 'real-world', 'single-modal', 'state-of-the-art', 'user-specific', 'long-form', 'long-term', 'multimodal', 'accurately'], 'year': 2025, 'citation_count': 148, 'quality_score': 90, 'quality_classification': 'Excellent'}

• {'title': 'A multi-agent system for automating scientific discovery', 'research_area': 'Multi-Agent Systems', 'research_problem': 'Scientific discovery is driven by the iterative process of observation, hypothesis generation, experimentation and data analysis.', 'methodology': ['Experimental Evaluation'], 'keywords': ['multi-agent', 'agent', 'age-related', 'knowledge', 'ripasudil-induced', 'semi-autonomous', 'robin', 'experimental', 'follow-up', 'hypothese', 'degeneration', 'experiment', 'generation', 'hypothesis', 'phagocytosis'], 'year': 2026, 'citation_count': 102, 'quality_score': 90, 'quality_classification': 'Excellent'}

Highest Cited Paper

title: Towards smart factory for industry 4.0: a self-organized multi-agent system with big data based feedback and coordination

authors: ['Shiyong Wang', 'J. Wan', 'Daqiang Zhang', 'Di Li', 'Chunhua Zhang']

year: 2016

citation_count: 1263

summary: Abstract not available for analysis.

research_problem: Not Available

methodology: ['Not Identified']

key_contributions: []

future_work: []

keywords: []

research_area: Multi-Agent Systems

paper_score: 50

paper_quality: Good

Latest Paper

title: A multi-agent system for automating scientific discovery

authors: ['A. Ghareeb', 'Benjamin Chang', 'L. Mitchener', 'Angela Yiu', 'Caralyn J. Szostkiewicz', 'D. Shved', 'Gavin Gyimesi', 'Jon M. Laurent', 'S. Wright', 'Muhammed Razzak', 'Andrew D. White', 'S. Finnemann', 'Michaela M. Hinks', 'Samuel G. Rodriques']

year: 2026

citation_count: 102

summary: Scientific discovery is driven by the iterative process of observation, hypothesis generation, experimentation and data analysis. Despite recent advancements in applying artificial intelligence (AI) to biology, no system has yet automated all these stages1, 2–3. Here we introduce Robin, a multi-agent system capable of fully automating both hypothesis generation and data analysis for experimental biology. By integrating literature search agents with data analysis agents, Robin can generate hypotheses, propose experiments, interpret experimental results and generate updated hypotheses, achieving a semi-autonomous approach to scientific discovery. By applying this system, we were able to identify promising therapeutic candidates for dry age-related macular degeneration, the major cause of blindness in the developed world4,5. Robin proposed enhancing retinal pigment epithelium phagocytosis as a therapeutic strategy, and identified and confirmed in vitro efficacy for ripasudil and KL001. Ripasudil is a clinically used Rho kinase inhibitor that, to our knowledge, has never previously been proposed for the treatment of dry age-related macular degeneration. To elucidate the mechanism of ripasudil-induced upregulation of phagocytosis, Robin then proposed and analysed a follow-up RNA sequencing experiment, which revealed upregulation of ABCA1, which encodes a lipid efflux pump and represents a possible novel target. All hypotheses, experimental directions, data analyses and research_problem: Scientific discovery is driven by the iterative process of observation, hypothesis generation, experimentation and data analysis.

methodology: ['Experimental Evaluation']

key_contributions: ['Here we introduce Robin, a multi-agent system capable of fully automating both hypothesis generation and data analysis for experimental biology.']

future_work: []

keywords: ['multi-agent', 'agent', 'age-related', 'knowledge', 'ripasudil-induced', 'semi-autonomous', 'robin', 'experimental', 'follow-up', 'hypothese', 'degeneration', 'experiment', 'generation', 'hypothesis', 'phagocytosis']

research_area: Multi-Agent Systems

paper_score: 90

paper_quality: Excellent

Common Methodologies

• Architecture

• Experimental Evaluation

• Framework

• Large Language Model

• Not Identified

• Retrieval-Augmented Generation

Research Areas

• Multi-Agent Systems

• Retrieval-Augmented Generation

Common Keywords

• accessible

• accurately

• achieve

• age-related

• agent

• agentic

• behavior

• benchmark

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- self-reflection
- semi-autonomous
- single-modal
- state-of-the-art
- tasks
- user-specific
- vision

Methodology Frequency

Experimental Evaluation: 4

Framework: 2

Architecture: 1

Large Language Model: 1

Not Identified: 1

Retrieval-Augmented Generation: 1

Most Common Methodology: Experimental Evaluation

Quality Distribution

Very Good: 2

Good: 1

Excellent: 2

Publication Years

2016: 1

2024: 2

2025: 1

2026: 1

Citation Statistics

highest: 1263

lowest: 98

average: 372.4

total: 1862

Comparison Summary

total_papers: 5

most_common_methodology: Experimental Evaluation

research_areas: ['Multi-Agent Systems', 'Retrieval-Augmented Generation']

quality_distribution: {'Very Good': 2, 'Good': 1, 'Excellent': 2}

citation_statistics: {'highest': 1263, 'lowest': 98, 'average': 372.4, 'total': 1862}

highest_cited_title: Towards smart factory for industry 4.0: a self-organized multi-agent system with big data based feedback and coordination

latest_paper_title: A multi-agent system for automating scientific discovery

Research Gap Analysis

Total Papers: 5

Research Areas

• Multi-Agent Systems
• Retrieval-Augmented Generation

Common Keywords

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Future Work

- We anticipate that our framework and observations will provide valuable insights for further research into the safety of multi-agent systems.

Research Area Distribution

Multi-Agent Systems: 4

Retrieval-Augmented Generation: 1

Keyword Frequency

agent: 4

multi-agent: 4

knowledge: 3

benchmark: 2

state-of-the-art: 2

accessible: 1

accurately: 1

achieve: 1

age-related: 1

agentic: 1

behavior: 1

collective: 1

concern: 1

dangerous: 1
dark: 1
degeneration: 1
experiment: 1
experimental: 1
follow-up: 1
generation: 1
high-performing: 1
high-resolution: 1
hypothese: 1
hypothesis: 1
intelligence: 1
long-form: 1
long-term: 1
magentic-one: 1
memory: 1
mirix: 1
multi-step: 1
multimodal: 1
open-source: 1
phagocytosis: 1
psychological: 1
psychology: 1
psysafe: 1
re-plan: 1
real-world: 1
reasoning: 1
ripasudil-induced: 1
risks: 1
robin: 1
safety: 1
self-reflection: 1
semi-autonomous: 1
single-modal: 1
tasks: 1
user-specific: 1
vision: 1

Research Trends

• agent

• multi-agent

• knowledge

• benchmark

• state-of-the-art

• accessible

• accurately

• achieve

• age-related

• agentic

• behavior

• collective

• concern

• dangerous

• dark

Gap Categories

datasets: []

models: ["Modern AI agents, driven by advances in large foundation models, promise to enhance our productivity and transform our lives by augmenting our knowledge and capabilities. To achieve this vision, AI agents must effectively plan, perform multi-step reasoning and actions, respond to novel observations, and recover from errors, to successfully complete complex tasks across a wide range of scenarios. In this work, we introduce Magentic-One, a high-performing open-source agentic system for solving such tasks. Magentic-One uses a multi-agent architecture where a lead agent, the Orchestrator, plans, tracks progress, and re-plans to recover from errors. Throughout task execution, the Orchestrator directs other specialized agents to perform tasks as needed, such as operating a web browser, navigating local files, or writing and executing Python code. We show that Magentic-One achieves statistically competitive performance to the state-of-the-art on three diverse and challenging agentic benchmarks: GAIA, AssistantBench, and WebArena. Magentic-One achieves these results without modification to core agent capabilities or to how they collaborate, demonstrating progress towards generalist agentic systems. Moreover, Magentic-One's modular design allows agents to be added or removed from the team without additional prompt tuning or training, easing development and making it extensible to future scenarios. We provide an open-source implementation of", 'Modern AI agents, driven by advances in large foundation models, promise to enhance our productivity and transform our lives by augmenting our knowledge and capabilities.', "Multi-agent systems, when enhanced with Large Language Models (LLMs), exhibit profound capabilities in collective intelligence. However, the potential misuse of this intelligence for malicious purposes presents significant risks. To date, comprehensive research on the safety issues associated with multi-agent systems remains limited. In this paper, we explore these concerns through the innovative lens of agent psychology, revealing that the dark psychological states of agents constitute a significant threat to safety. To tackle these concerns, we propose a comprehensive framework (PsySafe) grounded in agent psychology, focusing on three key areas: firstly, identifying how dark personality traits in agents can lead to risky behaviors; secondly, evaluating the safety of multi-agent systems from the psychological and behavioral perspectives, and thirdly, devising effective strategies to mitigate these risks. Our experiments reveal several intriguing phenomena, such as the collective dangerous behaviors among agents, agents' self-reflection when engaging in dangerous behavior, and the correlation between agents' psychological assessments and dangerous behaviors. We anticipate that our framework and observations will provide valuable insights for further research into the safety of multi-agent systems. We will make our data and code publicly accessible at https://github.com/AI4Good24/PsySafe.", 'Large Language Model', 'MIRIX: Multi-Agent Memory System for LLM-Based Agents', "Although memory capabilities of AI agents are gaining increasing attention, existing solutions remain fundamentally limited. Most rely on flat, narrowly scoped memory components, constraining their ability to personalize, abstract, and reliably recall user-specific information over time. To this end, we introduce MIRIX, a modular, multi-agent memory system that redefines the future of AI memory by solving the field's most critical challenge: enabling language models to truly remember. Unlike prior approaches, MIRIX transcends text to embrace rich visual and multimodal experiences, making memory genuinely useful in real-world scenarios. MIRIX consists of six distinct, carefully structured memory types: Core, Episodic, Semantic, Procedural, Resource Memory, and Knowledge Vault, coupled with a multi-agent framework that dynamically controls and coordinates updates and retrieval. This design enables agents to persist, reason over, and accurately retrieve diverse, long-term user data at scale. We validate MIRIX in two demanding settings. First, on ScreenshotVQA, a challenging multimodal benchmark comprising nearly 20,000 high-resolution computer screenshots per sequence, requiring deep contextual understanding and where no existing memory systems can be applied, MIRIX achieves 35% higher accuracy than the RAG baseline while reducing storage requirements by 99.9%. Second, on LOCOMO, a long-form conversation benchmark with single-modal textual input, MIRIX attains state-of-the-art performance", "To this end, we introduce MIRIX, a modular, multi-agent memory system that redefines the future of AI memory by solving the field's most critical challenge: enabling language models to truly remember."]

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hallucination: []

knowledge_freshness: []

evaluation: ["Modern AI agents, driven by advances in large foundation models, promise to enhance our productivity and transform our lives by augmenting our knowledge and capabilities. To achieve this vision, AI agents must effectively plan, perform multi-step reasoning and actions, respond to novel observations, and recover from errors, to successfully complete complex tasks across a wide range of scenarios. In this work, we introduce Magentic-One, a high-performing open-source agentic system for solving such tasks. Magentic-One uses a multi-agent architecture where a lead agent, the Orchestrator, plans, tracks progress, and re-plans to recover from errors. Throughout task execution, the Orchestrator directs other specialized agents to perform tasks as needed, such as operating a web browser, navigating local files, or writing and executing Python code. We show that Magentic-One achieves statistically competitive performance to the state-of-the-art on three diverse and challenging agentic benchmarks: GAIA, AssistantBench, and WebArena. Magentic-One achieves these results without modification to core agent capabilities or to how they collaborate, demonstrating progress towards generalist agentic systems. Moreover, Magentic-One's modular design allows agents to be added or removed from the team without additional prompt tuning or training, easing development and making it extensible to future scenarios. We provide an open-source implementation of", 'Experimental Evaluation', 'We show that Magentic-One achieves statistically competitive performance to the state-of-the-art on three diverse and challenging agentic benchmarks: GAIA, AssistantBench, and WebArena.', 'PsySafe: A Comprehensive Framework for Psychological-based Attack, Defense, and Evaluation of Multi-agent System Safety', "Although memory capabilities of AI agents are gaining increasing attention, existing solutions remain fundamentally limited. Most rely on flat, narrowly scoped memory components, constraining their ability to personalize, abstract, and reliably recall user-specific information over time. To this end, we introduce MIRIX, a modular, multi-agent memory system that redefines the future of AI memory by solving the field's most critical challenge: enabling language models to truly remember. Unlike prior approaches, MIRIX transcends text to embrace rich visual and multimodal experiences, making memory genuinely useful in real-world scenarios. MIRIX consists of six distinct, carefully structured memory types: Core, Episodic, Semantic, Procedural, Resource Memory, and Knowledge Vault, coupled with a multi-agent framework that dynamically controls and coordinates updates and retrieval. This design enables agents to persist, reason over, and accurately retrieve diverse, long-term user data at scale. We validate MIRIX in two demanding settings. First, on ScreenshotVQA, a challenging multimodal benchmark comprising nearly 20,000 high-resolution computer screenshots per sequence, requiring deep contextual understanding and where no existing memory systems can be applied, MIRIX achieves 35% higher accuracy than the RAG baseline while reducing storage requirements by 99.9%. Second, on LOCOMO, a long-form conversation benchmark with single-modal textual input, MIRIX attains state-of-the-art performance", 'First, on ScreenshotVQA, a challenging multimodal benchmark comprising nearly 20,000 high-resolution computer screenshots per sequence, requiring deep contextual understanding and where no existing memory systems can be applied, MIRIX achieves 35% higher accuracy than the RAG baseline while reducing storage requirements by 99.9%.']

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coordination: ['Magentic-One: A Generalist Multi-Agent System for Solving Complex Tasks', "Modern AI agents, driven by advances in large foundation models, promise to enhance our productivity and transform our lives by augmenting our knowledge and capabilities. To achieve this vision, AI agents must effectively plan, perform multi-step reasoning and actions, respond to novel observations, and recover from errors, to successfully complete complex tasks across a wide range of scenarios. In this work, we introduce Magentic-One, a high-performing open-source agentic system for solving such tasks. Magentic-One uses a multi-agent architecture where a lead agent, the Orchestrator, plans, tracks progress, and re-plans to recover from errors. Throughout task execution, the Orchestrator directs other specialized agents to perform tasks as needed, such as operating a web browser, navigating local files, or writing and executing Python code. 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We anticipate that our framework and observations will provide valuable insights for further research into the safety of multi-agent systems. We will make our data and code publicly accessible at https://github.com/AI4Good24/PsySafe.", 'To tackle these concerns, we propose a comprehensive framework (PsySafe) grounded in agent psychology, focusing on three key areas: firstly, identifying how dark personality traits in agents can lead to risky behaviors; secondly, evaluating the safety of multi-agent systems from the psychological and behavioral perspectives, and thirdly, devising effective strategies to mitigate these risks.', 'We anticipate that our framework and observations will provide valuable insights for further research into the safety of multi-agent systems.', 'Towards smart factory for industry 4.0: a self-organized multi-agent system with big data based feedback and coordination', 'MIRIX: Multi-Agent Memory System for LLM-Based Agents', "Although memory capabilities of AI agents are gaining increasing attention, existing solutions remain fundamentally limited. Most rely on flat, narrowly scoped memory components, constraining their ability to personalize, abstract, and reliably recall user-specific information over time. To this end, we introduce MIRIX, a modular, multi-agent memory system that redefines the future of AI memory by solving the field's most critical challenge: enabling language models to truly remember. Unlike prior approaches, MIRIX transcends text to embrace rich visual and multimodal experiences, making memory genuinely useful in real-world scenarios. MIRIX consists of six distinct, carefully structured memory types: Core, Episodic, Semantic, Procedural, Resource Memory, and Knowledge Vault, coupled with a multi-agent framework that dynamically controls and coordinates updates and retrieval. This design enables agents to persist, reason over, and accurately retrieve diverse, long-term user data at scale. We validate MIRIX in two demanding settings. First, on ScreenshotVQA, a challenging multimodal benchmark comprising nearly 20,000 high-resolution computer screenshots per sequence, requiring deep contextual understanding and where no existing memory systems can be applied, MIRIX achieves 35% higher accuracy than the RAG baseline while reducing storage requirements by 99.9%. Second, on LOCOMO, a long-form conversation benchmark with single-modal textual input, MIRIX attains state-of-the-art performance", "To this end, we introduce MIRIX, a modular, multi-agent memory system that redefines the future of AI memory by solving the field's most critical challenge: enabling language models to truly remember.", 'A multi-agent system for automating scientific discovery', 'Scientific discovery is driven by the iterative process of observation, hypothesis generation, experimentation and data analysis. Despite recent advancements in applying artificial intelligence (AI) to biology, no system has yet automated all these stages1, 2–3. Here we introduce Robin, a multi-agent system capable of fully automating both hypothesis generation and data analysis for experimental biology. By integrating literature search agents with data analysis agents, Robin can generate hypotheses, propose experiments, interpret experimental results and generate updated hypotheses, achieving a semi-autonomous approach to scientific discovery. By applying this system, we were able to identify promising therapeutic candidates for dry age-related macular degeneration, the major cause of blindness in the developed world4,5. Robin proposed enhancing retinal pigment epithelium phagocytosis as a therapeutic strategy, and identified and confirmed in vitro efficacy for ripasudil and KL001. Ripasudil is a clinically used Rho kinase inhibitor that, to our knowledge, has never previously been proposed for the treatment of dry age-related macular degeneration. To elucidate the mechanism of ripasudil-induced upregulation of phagocytosis, Robin then proposed and analysed a follow-up RNA sequencing experiment, which revealed upregulation of ABCA1, which encodes a lipid efflux pump and represents a possible novel target. All hypotheses, experimental directions, data analyses and', 'Here we introduce Robin, a multi-agent system capable of fully automating both hypothesis generation and data analysis for experimental biology.']

security: ['PsySafe: A Comprehensive Framework for Psychological-based Attack, Defense, and Evaluation of Multi-agent System Safety']

scalability: ["Although memory capabilities of AI agents are gaining increasing attention, existing solutions remain fundamentally limited. Most rely on flat, narrowly scoped memory components, constraining their ability to personalize, abstract, and reliably recall user-specific information over time. To this end, we introduce MIRIX, a modular, multi-agent memory system that redefines the future of AI memory by solving the field's most critical challenge: enabling language models to truly remember. Unlike prior approaches, MIRIX transcends text to embrace rich visual and multimodal experiences, making memory genuinely useful in real-world scenarios. MIRIX consists of six distinct, carefully structured memory types: Core, Episodic, Semantic, Procedural, Resource Memory, and Knowledge Vault, coupled with a multi-agent framework that dynamically controls and coordinates updates and retrieval. This design enables agents to persist, reason over, and accurately retrieve diverse, long-term user data at scale. We validate MIRIX in two demanding settings. First, on ScreenshotVQA, a challenging multimodal benchmark comprising nearly 20,000 high-resolution computer screenshots per sequence, requiring deep contextual understanding and where no existing memory systems can be applied, MIRIX achieves 35% higher accuracy than the RAG baseline while reducing storage requirements by 99.9%. Second, on LOCOMO, a long-form conversation benchmark with single-modal textual input, MIRIX attains state-of-the-art performance"]

explainability: []

reasoning: []

memory: []

planning: []

general: ['Although memory capabilities of AI agents are gaining increasing attention, existing solutions remain fundamentally limited.']

Research Gaps

• Retrieval quality and the ability to consistently retrieve relevant information for complex queries remain important research challenges.

• Improved grounding, factuality, and reliable source attribution are required to reduce unsupported or incorrect generated information.

• More robust and standardized evaluation and benchmarking methods are required.

• Current models and architectures still require improvements in reliability, adaptability, and generalization.

• Reliable coordination and communication between multiple agents remains an open research challenge.

• Security, privacy, trust, and robustness remain important unresolved challenges.

• Scalability, computational cost, and efficient resource usage remain challenges for practical deployment.

• Further validation in real-world applications and deployment environments is required.

• We show that Magentic-One achieves statistically competitive performance to the state-of-the-art on three diverse and challenging agentic benchmarks: GAIA, AssistantBench, and WebArena.

• Experimental Evaluation

Emerging Topics

• agent

• multi-agent

• knowledge

• benchmark

• state-of-the-art

Recommendations

• Explore more reliable autonomous and multi-agent systems with improved coordination and task execution.

• Investigate improved inter-agent communication and coordination mechanisms for reliable multi-agent task execution.

• Improve retrieval quality and document selection for complex research queries.

• Strengthen source grounding and citation verification to improve factual reliability.

• Evaluate proposed approaches using standardized benchmarks and reproducible evaluation protocols.

• Compare stronger model architectures and established baselines to identify improvements in reliability and generalization.

• Investigate stronger security, privacy, trust, and adversarial robustness mechanisms.

• Optimize computational cost and resource usage to improve scalability for practical deployment.

• Validate the proposed approaches in realistic real-world environments to assess scalability and practical usability.

• Design more robust, modular, and scalable AI frameworks for practical research applications.

Summary

dominant_area: Multi-Agent Systems

top_trend: agent

future_work_items: 1

research_gap_count: 10

Experiment Plan

Research Areas

- Multi-Agent Systems
- Retrieval-Augmented Generation

Recommended Datasets

- AgentBench
- BEIR
- GAIA Benchmark
- HotpotQA
- MS MARCO
- Multi-Agent Benchmark
- Natural Questions
- OpenAI Evals
- TriviaQA

Baseline Models

- AutoGen
- BM25 Retrieval
- CrewAI
- Dense Retrieval RAG
- Hybrid Search RAG
- LangGraph
- Naive RAG
- ReAct
- Single-Agent LLM
- Vanilla RAG

Evaluation Metrics

- Accuracy
- Precision

• Recall
• F1-Score
• Retrieval Precision
• Retrieval Recall
• Context Relevance
• Answer Relevance
• Faithfulness
• Groundedness
• Task Success Rate
• Planning Accuracy
• Tool-Use Accuracy
• Agent Coordination Score
• ROC-AUC

Hardware Requirements

cpu: Intel Core i7 / AMD Ryzen 7

ram: 32 GB

gpu: NVIDIA RTX 3060 / RTX 4060 or better

storage: 200 GB SSD

Validation Strategy

• Train-Test Split
• K-Fold Cross Validation
• Hyperparameter Tuning
• Statistical Significance Testing
• Retrieval Evaluation
• Answer Quality Evaluation
• Task Success Evaluation
• Agent Coordination Evaluation
• Tool-Use Evaluation

Experimental Workflow

• Collect Knowledge Documents
• Preprocess and Chunk Documents
• Build Document Index
• Configure Retrieval System
• Retrieve Relevant Context
• Generate Grounded Response
• Evaluate Retrieval Quality

• Evaluate Answer Quality

• Compare Baseline and Proposed RAG

• Analyze Errors

• Draw Conclusions

Gap Based Recommendations

• Compare BM25, dense retrieval, and hybrid retrieval methods.

• Evaluate source attribution, faithfulness, and groundedness.

• Use standardized benchmarks and multiple evaluation metrics.

• Compare centralized and decentralized multi-agent coordination strategies.

• Evaluate robustness against prompt injection, adversarial inputs, and unauthorized tool usage.

• Measure computational cost, latency, resource consumption, and scalability.

• Validate the system in realistic real-world deployment scenarios.

Report Summary

Dominant Research Area: Multi-Agent Systems

Top Research Trend: agent

Highest Cited Paper: Towards smart factory for industry 4.0: a self-organized multi-agent system with big data based feedback and coordination

Latest Paper: A multi-agent system for automating scientific discovery

Recommended Datasets

• AgentBench

• BEIR

• GAIA Benchmark

• HotpotQA

• MS MARCO

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• Natural Questions

• OpenAI Evals

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• Groundedness
• Task Success Rate
• Planning Accuracy
• Tool-Use Accuracy
• Agent Coordination Score
• ROC-AUC

Conclusion

The report has been generated using a personalized multi-agent pipeline. The explanation style was adapted according to the Intermediate profile to improve readability and usefulness.